CS4480 UK Price Paid — project pack (all-in-one)

This zip includes code + cleaned data + report figures.

What works out of the box	What needs extra setup
Computer: install packages → run figure / analysis scripts	Hadoop VM: needs the VirtualBox shared-folder layout (see §5)

Local paths match this pack layout. VM commands match the Bitnami Hadoop scripts, but only succeed if the mount layout in §5 is correct.

1. Unzip (Windows)

1. Right-click CS4480_UK_PricePaid_source.zip → Extract All…
2. Extract to e.g. Desktop\CS4480_UK_PricePaid_source
3. Open that folder in File Explorer

Name	What it is
README.pdf	This guide
requirements.txt	Python packages
local_python\	Scripts for your computer
parallel\	Hadoop / Spark / MR scripts (folder name must stay parallel)
figures4papers\	Report PNGs (fig0–fig13)
data\processed\	Trend / k-means / rules / timing CSVs
data\parallel\	Feature matrices + centroids

Quick check (File Explorer)

Confirm these files exist:

Check	Path inside the project folder
1	data\processed\uk_monthly_by_type.csv
2	data\parallel\init_centroids_k4.txt
3	data\parallel\uk_features_scaled_2018plus_sample10pct.txt

2. Computer setup

Open Command Prompt or PowerShell:

```
cd Desktop\CS4480_UK_PricePaid_source
python -m pip install -r requirements.txt
```

(Change cd if you extracted elsewhere.)

3. Reproduce on your computer (recommended)

Always run from the project root (the folder that contains local_python and data).

Fastest — refresh report figures

```
python local_python\make_figures4papers.py
python local_python\make_pipeline_flowchart.py
```

Re-run analyses

```
python local_python\analyze_trends_kmeans.py
python local_python\analyze_rule_mining.py
python local_python\make_figures4papers.py
```

Rebuild from raw (optional)

Put pp-complete.csv (~3 GB) in the project root, then:

```
python local_python\build_uk_panel.py
```

4. What's already in data\

data\processed\

Panels, k-means tables, association rules, timing CSVs for fig9–fig13.

data\parallel\

File	Role
init_centroids_k4.txt	Shared k=4 init
uk_features_scaled_2018plus_sample10pct.txt	Local k-means / VM mid-size (~58 MB)
uk_features_scaled_2018plus.txt	Full 2018+ scaled (~570 MB)
uk_features_raw_2018plus.txt	Rule mining (~580 MB)

Not included: pp-complete.csv, scale-up matrices uk_features_scaled_scale_n*.txt.

5. Hadoop VM — read carefully

Scripts hard-code this layout on the VM:

```
/home/bitnami/uk_housing/          ← shared folder mount point
  data/
    processed/
    parallel/                      ← feature files (sample10pct, centroids, …)
  parallel/                        ← code from this zip's parallel\ folder
    env.sh
    run_benchmark.sh
    spark_kmeans_uk.scala
    mr_kmeans/
    ...
```

Common mistakes

Mistake	What happens
Only copy scripts, forget data\ on the VM	Spark/MR: file not found
Share the zip folder but rename parallel\	Scripts look for .../parallel/env.sh → fail
Shared folder not mounted at /home/bitnami/uk_housing	All paths break

Windows CRLF line endings on .sh	bash errors like $'\r': command not found
HDFS/YARN not running	hdfs / MapReduce fail

Required setup (Bitnami Hadoop VirtualBox VM)

1. On the Windows host, keep this unzipped project as the shared-folder content

(VirtualBox shared folder name is typically housing → mounts to /home/bitnami/uk_housing).

2. On the VM, confirm both exist:

```
ls /home/bitnami/uk_housing/data/parallel/init_centroids_k4.txt
ls /home/bitnami/uk_housing/parallel/env.sh
```

If data/... is missing, the shared folder is not pointing at this unzipped project.

If parallel/env.sh is missing, the host folder was not named parallel (do not rename it).

3. Fix line endings + permissions (do this once):

```
cd /home/bitnami/uk_housing/parallel
sed -i 's/\r$//' *.sh mr_kmeans/*.sh
chmod +x *.sh mr_kmeans/*.sh mr_kmeans/*.py
```

4. Start services and run mid-size benchmark:

```
source /home/bitnami/uk_housing/parallel/env.sh
bash /home/bitnami/uk_housing/parallel/start_services.sh
bash /home/bitnami/uk_housing/parallel/run_benchmark.sh
```

This runs Spark + MapReduce on uk_features_scaled_2018plus_sample10pct.txt and writes logs under data/processed/.

If VM setup is unfamiliar

Skip §5. Figures + analyses on your computer (§3) are enough for most report work. Parallel timing numbers/figures are already in figures4papers\ and data\processed\.

6. Script map

Script (local_python\)	Purpose
build_uk_panel.py	Raw → panels + features
analyze_trends_kmeans.py	Trends + sklearn k-means
analyze_rule_mining.py	Apriori rules
make_figures4papers.py	fig1–fig13
make_pipeline_flowchart.py	fig0

Script (parallel\)	Purpose
env.sh / start_services.sh	Env + HDFS/YARN
run_benchmark.sh	Spark + MR mid-size timing
spark_kmeans_uk.scala	Spark k-means
python_kmeans_uk.py	ProcessPool baseline
mr_kmeans*	Hadoop Streaming k-means

Property types: D=Detached, S=Semi-detached, T=Terraced, F=Flat/maisonette, O=Other.